

A Soft Adaptive Gripper for Non-Cooperative Object Grasping

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Abstract—The capture and manipulation of non-cooperative objects remains a key challenge in robotic manipulation, particularly in the unstructured environment of space. While adept at industrial gripping tasks, rigid grippers struggle to grasp objects when they lack precise positional data about those objects. A largely unexplored yet promising solution to these limitations can be found in compliant or soft grippers, which physically adapt to the geometry of a captured object. This grasp modality allows for the capture of objects with complex geometry without full sensory information. This work explores the use of a biologically inspired soft robotic gripper, designed to conduct ground-based experiments to evaluate its potential for non-destructive capture and manipulation, as well as its viability for subsequent deployment in space operations.

The digits of this soft robotic gripper are modeled after the flexible paddle-like shape of cephalopod tentacles, which are used to capture evasive prey. The gripper's 3D-printed fingers are designed to consistently capture objects of varying geometry, as well as objects in motion relative to the gripper. The flexibility of the fingers allows for adaptable gripping that conforms around these objects without causing damage.

In this paper we detail the design, development, implementation, and evaluation of the soft gripper, mounted to a seven degree of freedom robotic arm. We detail the methodology behind the tendon-driven finger design and the construction of several prototypes. In addition, we provide qualitative and quantitative experimental analyses from capture testing. These findings demonstrate our soft gripper's reliability and reduced risk of target damage, providing a superior alternative to rigid grippers for capturing objects in unstructured space environments.

I. INTRODUCTION

Earth's orbit is becoming increasingly cluttered with space debris, posing a threat to future launches and active satellite operation. These human-made objects vary in size from defunct satellites and rocket bodies to untrackable pieces of shrapnel. According to the European Space Agency, there are estimated to be 54,000 space objects greater than 10 cm, and 1.2 million space debris objects between 1 cm to 10 cm. [1] With globally increased in-space operations contributing to additional space debris, reliable collection, capture, and manipulation of these objects remains a challenge. This issue is further exacerbated by the need to capture some targets non-destructively, as a forceful capture could damage fragile components or create additional fragmentation.

Current debris removal technologies rely on direct satellite acquisition. These methods include nets, harpoons, or direct capture with robotic manipulators; however, each of these

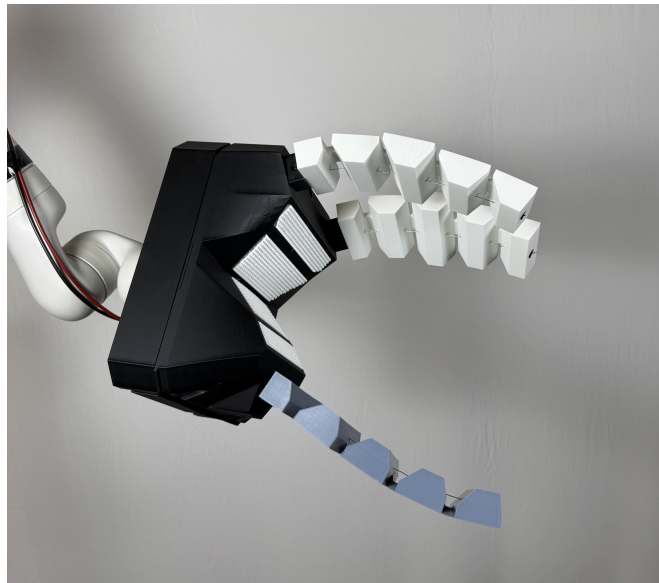


Fig. 1. Robotic gripper with soft digits

methods faces significant limitations when dealing with the unpredictability of space debris. Several experimental missions have attempted to address these challenges, most notably RemoveDEBRIS, which demonstrated the feasibility of net and harpoon systems [2] [3]. While effective in concept, both these methods pose additional risk. Net capture can entangle a target, but imparted forces can result in unpredictable changes in the target's velocity, and this method lacks manipulability of the captured object [2]. Similarly, harpoon systems risk destroying fragile components and fragmenting larger targets, creating additional debris. In a similar vein, traditional grippers are unreliable for capturing objects with complex and fragile geometry due to their dependence on precise target pose information, which can be difficult to acquire [4]. Given these limitations, the capture and manipulation of space debris remains an active research area.

An alternative promising solution lies in soft robotics, which are made from deformable material that can adapt to the complex shape of targeted objects, without requiring exact knowledge of the target's pose. This inherent flexibility allows these grippers to envelop a captured object, creating a non-destructive grasp [5]. Likewise, underactuated grippers, which use redundant joints to passively adapt to targets have been developed for use in unstructured environments [6] [7] [8]. These types of grippers have been shown to be an option, even in the harsh environment of space [9], [10].

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A cable-driven soft gripper was successfully developed for the acquisition of space debris in [11], showing that a soft web like structure can successfully capture non-cooperative objects. One drawback to this design is that objects captured by this gripper cannot be interacted with, as the solid web impedes any tool access to the target. This approach is not suitable for operations where the goal is to capture a target and perform maintenance or extract components.

This paper is organized as follows: Section II discusses the motivation for nature-inspired compliant grippers in the capture and manipulation of non-compliant objects. In Section III we detail the design and development of one such novel gripper, and Sections IV and V evaluate the performance of the prototype. Future work, discussion, and conclusions are provided in Sections VI and VII.

II. BACKGROUND AND MOTIVATION

Conventional robotic grippers are well suited for controlled environments, including pick and place tasks, assembly, and industrial applications. However robotic grasping in unstructured environments presents a significant challenge for these kinds of grippers. This is particularly true when handling uncooperative objects which can have properties characterized by deformability, fragility, asymmetric properties, or unknown pose. Manipulating these objects is outside the capabilities of traditional grippers that employ end effectors such as parallel-jaw grippers, which depend on precise knowledge about the object, including position and orientation. These systems often fail to account for the complex geometry of uncooperative objects, demonstrating a need for reliable manipulation of such targets.

Soft robotic grippers show promise as a different approach to solving these kinds of problems by using compliant materials that adapt to the complex shapes of uncooperative objects, while providing a gentle and secure grasp [12]. These soft manipulators are often inspired by biological systems, such as elephant trunks or cephalopod tentacles, and can manipulate objects without requiring the precise knowledge that conventional grippers do, making them ideal for use in unstructured environments. Pneumatically operated soft grippers have been shown to be adept at capturing and manipulating non-cooperative objects by providing high adaptability, but are limited by bulky infrastructure [13]. This highlights a need for an alternative system for applications where available space or weight is constrained.

The primary objective of this paper is to demonstrate the ability of a soft gripper to capture and manipulate non-cooperative objects, while maintaining a mobile platform that can be deployed in various harsh environments. This paper specifically addresses the challenge of capturing and grasping objects with complex geometries and unknown location, demonstrating that soft robotics offer a promising alternative to conventional approaches.

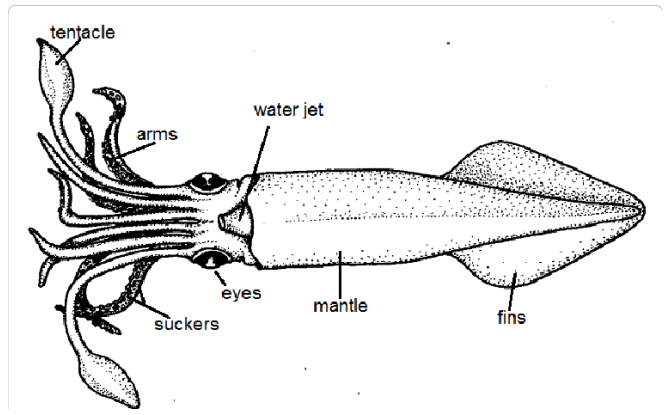


Fig. 2. Diagram of a cephalopod. Image Credit [22]

III. DESIGN AND METHODOLOGY

A. Overview

Nature often serves as motivation for the design of soft robotics, including trunks, tentacles, and other continuous shapes that occur naturally [14]. The gripper introduced in this paper is loosely based on the paddle like shape found in cephalopod tentacles (Figure 2). This structure is slender with a skinny base, a wider midsection, and a taper towards the end. Squids use this tentacle to curl around objects with maximum surface area, creating a secure grip on irregularly shaped objects. This gripper uses three flexible paddle-shaped digits, a rigid base that acts as a palm, and a compact tendon actuation system to effectively capture and manipulate non-conventional objects.

B. Material Selection

The fingers are made from Shore-95A TPU, chosen due to its high flexibility, while maintaining durability and being lightweight. This material allows the fingers to bend and conform to objects without damaging the object or the gripper itself. Similar gripper tools fabricated from a single continuous material have been implemented in [15]. TPU's high elasticity and friction properties make it ideal for soft robotic grippers, and similar materials have been used before in [16] [17], [9]. Due to its resistance to abrasion, cutting, tearing, and chemical corrosion, TPU is an optimal choice for harsh environments. It was observed during testing that the fingers were able to be repeatedly actuated, without any deformation or damage. The high coefficient of friction results in a grasp that can handle irregular surfaces. In contrast to this flexibility, the palm base was printed from Polylactic Acid (PLA) due to its rigidity and ease of use, creating a base to hold all electronic components.

C. Gripper Design

Two different paddle-shaped finger sizes are used - a main digit that is larger, with a contact area of 119.7 cm², and two smaller auxiliary fingers, each with a contact area of 76.6 cm², arranged around the palm. The auxiliary fingers are placed 135 degrees away from the main finger, and 90 degrees apart from one another. These secondary fingers are

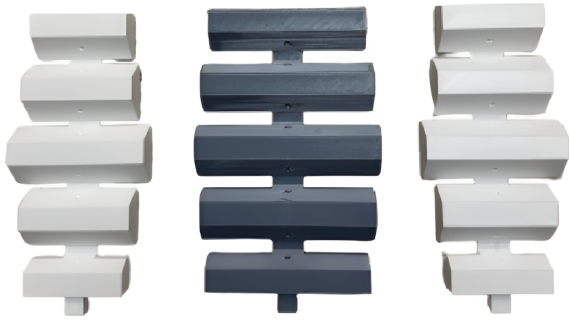


Fig. 3. Shape of main (center) and auxiliary fingers.

angled 10 degrees away from the midpoint of the palm, increasing the range of objects that can be grasped. This finger arrangement was found through trial and error with the described configuration being able to grasp objects of various sizes. The fingers are intentionally not placed equidistant around the palm to allow the gripper to grasp objects that are long and narrow, which extend past the edges of the palm. Each finger has 4 elastomer flexure joints that allow for the return to the open position when no cable tension is applied. The joints articulate through an arc of 70 degrees, resulting in a total finger flexion of 280 degrees when fully actuated. The contact face of each of the fingers is flat, maximizing surface area while also distributing forces to ensure that captured objects are not damaged. Polygonal side profiles allow for efficient bending while maintaining a constant structure. The fingers are printed with 25% infill, providing stability and durability while preserving flexibility. The primary finger initially captures an object, while the secondary fingers enhance stability by wrapping around the target.

The palm utilizes a trough-shaped design to passively align objects into an optimal capture configuration. This process maximizes contact area between the palm and the fingers, mitigating the potential for slippage of grasped targets. The palm's angle transitions from 110 degrees to 90 degrees at its center, compensating for initial misalignment. TPU inserts are placed along the sides of the palm to increase frictional forces on the target.

D. Actuation and Control

The gripper is actuated by a tendon driven system that consists of 22 gauge wound steel wire as tendons, routed internally through channels within each finger to a high-torque Pololu 37D DC motors located below each digit. A similar steel tendon and motor design was implemented in [18]. The tendons are terminated at the tip of each finger, where they are fixed in place. This multi-motor design allows for independent finger flexure control, aiding with capture of non-uniform objects. The tendon of each digit is run over a load cell sensor, providing feedback on the tension in that specific finger. This reading can be used to calculate the total tension in the tendons during capture.

E. Electronics and Communication

The gripper is interfaced with the Kinova Gen3 7 DOF robotic arm using a custom adapter plate to align with the bolt pattern on the arm's end. This arm provides a flexible testing platform to evaluate the performance of the gripper. Control signals are sent to the end effector via an internal microcontroller inside the Kinova arm. An Ethernet connection is used along with the Kinova Kortex application programming interface (API) to communicate with an Arduino ESP32 mounted inside the base of the gripper. This communication is done using Inter-Integrated-Circuit (I2C) protocol. The ESP32 microcontroller transmits position commands to two Roboclaw 2x7 controllers, which directly control the DC motor. Each motor is equipped with an encoder, and has custom tuned PID parameters for optimal control. The controllers are powered by the internally routed 24V supply, while the motors are powered via an external 24V power supply, due to their increased current draw.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

To assess the effectiveness of the gripper, a series of tests were conducted in which compliant and non-compliant objects were grasped. The goal of these tests was to qualitatively examine the performance of the gripper, specifically the shape and flexibility of the fingers, the size of the gripper, the angle and orientation of the fingers, the design of the palm, the strength of the motors, and the gripper's overall ability to capture and manipulate objects.

The experiments were conducted in a laboratory setting that was designed to simulate unstructured environments. The Kinova arm allowed the gripper to approach objects from multiple angles, and the wrist joint was used to position the fingers optimally to capture objects. Uncooperative objects that were tested include a 10x10x30cm cuboid, balloons of 10cm and 20cm radius, and a plush bird. These objects were chosen to showcase the gripper's ability to capture both co-operative and non-cooperative objects of varying symmetry without exact information about the objects location.

First, a baseline was taken to evaluate the gripper's performance in an unloaded state. This included measuring the speed of closure, as well as the force required to fully close the gripper. The closure speed was recorded to be 2.82 seconds. The motors were then actuated into a position where the gripper was observed to be closed, and the load cell readings were recorded.

Next, a series of static tests were conducted in which the objects detailed above were placed at a measured distance from the center of the palm and the gripper was actuated. The outcome of each of these tasks was recorded as a success (The gripper having full control and being able to manipulate the captured object) or failure. After establishing control of the target, the load cell readings were recorded to find the cable tension required to hold the object. This procedure was repeated with variations in the target objects orientation and distance from the palm (see Table I, II, III). These tests showed that the gripper was able to envelop and control objects that were placed at various locations

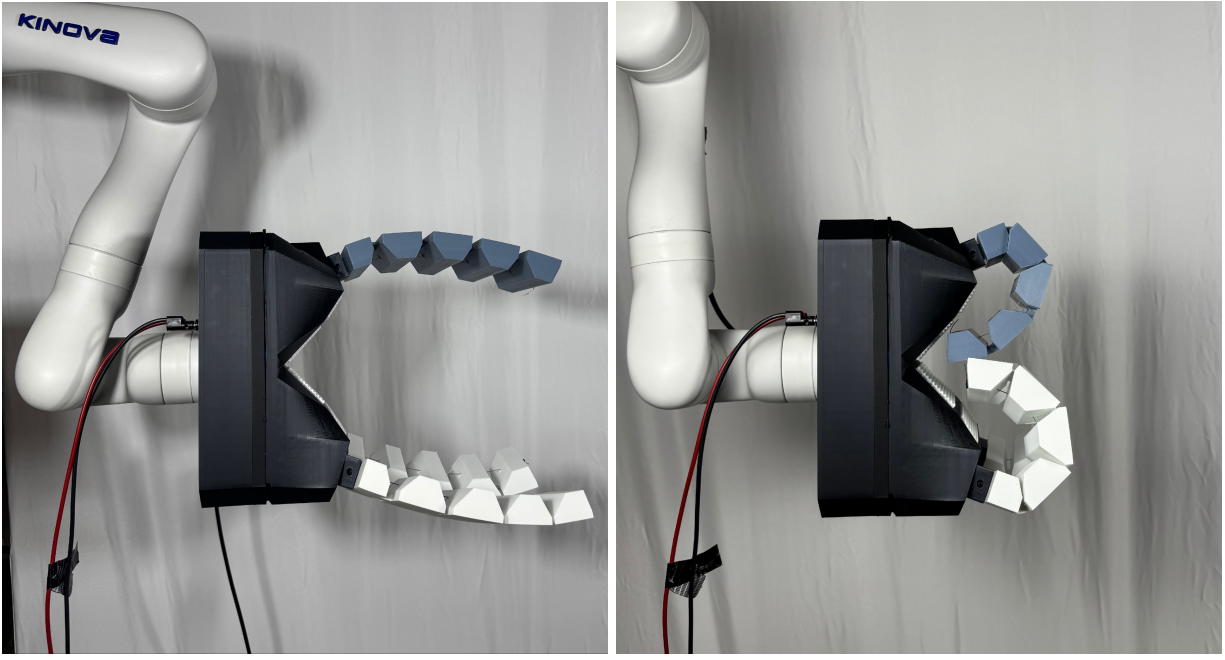


Fig. 4. Gripper in the open (default) and closed (actuated) configurations

inside the gripper, showing the ability to overcome attitude uncertainty between the target and the gripper pose.

To simulate the unpredictable conditions in space, a third test was conducted in which objects were suspended from above and given a random rotational velocity. The gripper was then maneuvered along a pre-planned trajectory to intercept the target, and the gripper was actuated. This test was conducted for 5 trials with each object, and the outcome of each trial was recorded (see Table IV).

Finally, to measure disturbance rejection the cuboid was placed vertically into an optimal grasping position. Once constrained, known weights were hung from the object until the target slipped from its original position. This experiment examined the gripper's ability to adapt to external disturbances applied to the object. The cuboid began to slip when an external force of 46.29N was applied.

V. DISCUSSION OF RESULTS

The baseline tests showed that the gripper can be actuated with sufficient speed to capture objects that have high unmeasured relative velocity, while the load cells allow for delicate handling where required. The required tension to close the gripper is well below the stall torque specified by the motors, leaving headroom for additional grasping force.

Static grasping tasks revealed that the gripper can mitigate relatively large positional uncertainties in the targets pose, making this gripper adept at grasping in low confidence environments. Any target placed within the gripper's capture area resulted in a successful capture, demonstrating the effectiveness of the design. The chosen finger orientation creates a large cone in which objects are pressed into the palm of the gripper, creating a secure hold. From the recorded forces required to capture each object (Table I, II, III) we

can observe that less force is required for more deformable objects (i.e., the plush bird and the balloons). This is due to the fingers ability to conform to the objects, maximizing surface area during contact. This provides a strong grasp with minimal contact forces, reducing the risk of damaging fragile components.

Trial	Distance to Palm(cm)	Rotation	Outcome	Total Cable Tension(N)
1	15	0	S	75.92
2	20	0	S	47.93
3	20	45	S	61.21
4	20	45	S	67.02
5	25	0	S	94.88

TABLE I

TRIAL RESULTS WITH 10CM X 10CM X 30CM CUBOID.

Trial	Distance to Palm(cm)	Balloon Radius(cm)	Outcome	Total Cable Tension(N)
1	15	20	S	17.53
2	20	20	S	11.20
3	20	10	S	12.04
4	20	10	S	14.25
5	25	10	S	22.99

TABLE II

TRIAL RESULTS WITH BALLOONS OF RADIUS 10CM AND 20CM.

The experimental tests in which the target was given a random rotational velocity demonstrate that the gripper is capable of capturing various tumbling payloads. This capability addresses a key limitation of conventional grippers, due to their reliance on precise pose knowledge [19].

The final experiments which involved applying known

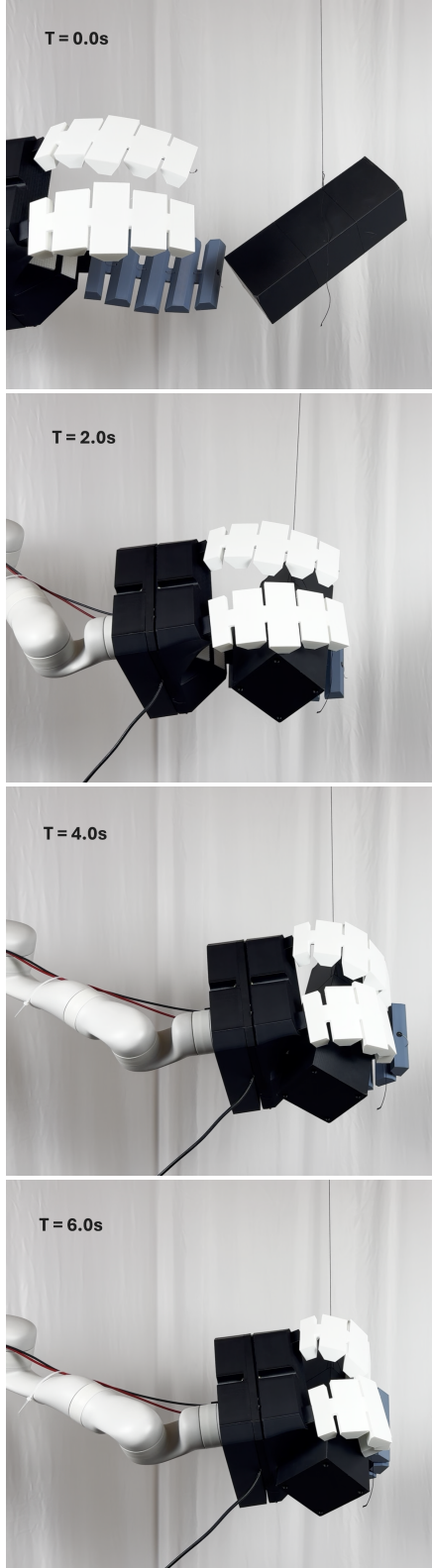


Fig. 5. Time lapse for capture of cuboid initially spinning at 142.1°/s .

Trial	Distance to Palm(cm)	Rotation	Outcome	Total Cable Tension(N)
1	15	0	S	81.28
2	20	0	S	30.34
3	20	90	S	33.17
4	20	180	S	35.75
5	25	0	S	26.85

TABLE III
TRIAL RESULTS WITH PLUSH BIRD.

Result	Rotational Velocity(°/s)
S	154.3
S	166.16
S	276.92
F	385.61
S	450.00

Result	Rotational Velocity(°/s)
S	62.83
F	73.31
S	96.26
S	142.1
S	143.43

TABLE IV
TRIAL RESULTS WITH INITIAL ROTATIONAL VELOCITY FOR PLUSH BIRD (LEFT) AND CUBOID (RIGHT).

weights to a constrained object, exhibited adequate disturbance rejection capability. This test quantifiably showed the performance of the gripper under external forces applied to the payload, which is a crucial factor in the control of objects in space. This is a direct result of the compliant palm shape and the fingers enclosing the object, creating a secure grip. The results of this test, along with the high grasping confidence, make this gripper suitable for object capture and manipulation in the dynamic and unpredictable environment of space. During multiple tests it was noted that the gripper successfully captured the target using unanticipated gripping geometry. Due to the flexibility of the fingers, objects could be leveraged against opposing fingers, rather than fully encompassed (See Figure 7). This was shown to be just as effective as a conventional grip in the disturbance rejection testing, but allowed for additional access to the captured object. This ability to leverage objects using partial grips significantly increases the gripper's versatility, further demonstrating the gripper's adaptability.

Overall, the gripper demonstrated notable adaptability in conforming to the shapes of tested objects. When faced with smaller objects, the fingers enveloped the target, adjusting to its geometry to maximize surface area. This was observed to create a firm grasp of small objects, allowing for manipulation. The mechanism demonstrated adequate ability to capture, manipulate, and release objects of varying sizes without damaging them. Despite unknown initial orientations and unexpected movements at the moment of contact, the gripper was able to capture all objects it was presented with.

These results showcase the effectiveness of the design decisions that were made during the design phase of this prototype. The fingers elasticity and friction demonstrated the ability of the gripper to conform to objects of various size while applying gentle pressure that resulted in a firm grasp, and the orientation of the fingers allowed the gripper to capture both small and large objects.

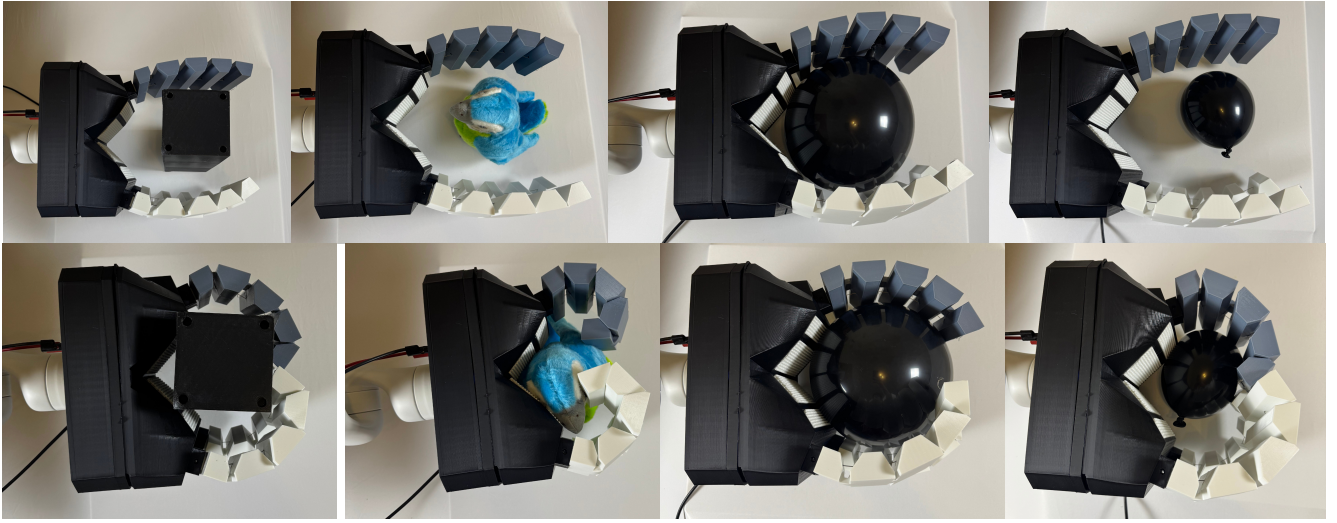


Fig. 6. Open (top row) and closed (bottom row) gripper configurations for each object captured during static grip test.

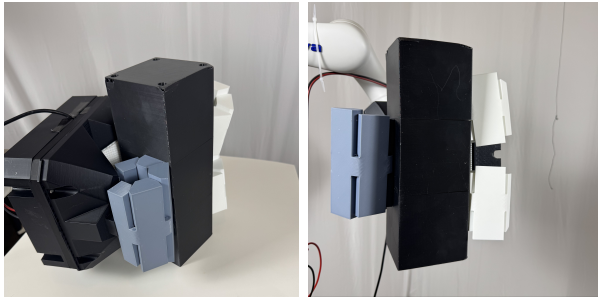


Fig. 7. Irregular grasping configuration during capture of cuboid.

VI. FUTURE WORK

Following the successful prototype of the tendon-driven soft gripper, future work will focus on improving robustness and capability. Material choice is a critical issue for deployment in the harsh conditions of space, and a suitable space-grade alternative must be found for the materials used in this prototype. To further document this gripper's performance in the vacuum of space, additional experiments can be done in a thermal-vacuum chamber. A camera sensor will be added to the center of the palm to monitor the position and orientation of the object during the capturing process. Vision-based sensing has been implemented previously to improve grasp planning in robotic manipulation [20]. The addition of this vision system will allow the system to autonomously track and capture a target without teleoperation or pre-planned trajectories. Along with this system, the use of an air-bearing table will simulate a more accurate microgravity environment for extended testing of the gripper without a human operator.

Simulation will be developed to better understand the gripper's behavior when faced with various conditions, including different object geometries, weights, and external forces. This will help development of optimal control algorithms with independent finger actuation. Simulations have been used previously to aid in optimizing control strategies for

soft robotic systems in [21].

VII. SUMMARY AND CONCLUSIONS

This paper describes the development and implementation of a novel tendon driven soft robotic gripper, and demonstrated its ability to adapt and securely grasp uncooperative objects with complex geometries and unknown positional data. This gripper was constructed leveraging the elastic properties of TPU to create a living hinge with sufficient rebound. A tendon based closure system proved to be an effective method of controlling the position of the fingers, offering an alternative to non-mobile pneumatically operated soft robots.

This design offered several key advantages over alternative space debris collection techniques: (1) the passively compliant digits can conform to the target, creating a non-destructive grasp for fragile and non-cooperative objects; (2) a large region of capture mitigates variation in the position and orientation of the target relative to the gripper, allowing for successful capture of objects with high sensing uncertainty; (3) the tendon-driven actuation system, utilizing a separate motor paired with a load cell for each finger, provides precise control of force applied to the target. This enables the gripper to achieve a tailored, adaptive grasp for each object; (4) the gripper's open-finger design allows for direct access to the captured object, facilitating post-capture manipulation. This capability distinguishes this gripper from other approaches that fully encompass the target, such as nets, which do not allow for subsequent payload manipulation.

VIII. ACKNOWLEDGEMENTS

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